

Yash Vishe

+1 (858)3965196 | yvishe@ucsd.edu | [linkedin.com/in/yashvishe](https://www.linkedin.com/in/yashvishe) | github.com/Yashvishe13

EDUCATION

University of California - San Diego, MS in Data Science, CA
University of Mumbai, BE in Computer Engineering, India

Sep. 2024 – Mar. 2026
Aug. 2019 – May. 2023

EXPERIENCE

AI Engineer, Vyasa AI, CA Apr. 2026 – Present

- Improved agent accuracy by 34% on internal benchmarks by developing behaviour-based **context graph** that enables a **self-evolving agent** to learn from experience and annotate successful and failed trajectories.
- Designed an agentic orchestration layer, unifying context across 10+ enterprise MCP servers through a shared context graph.

Machine Learning Researcher, UC San Diego, CA Oct. 2024 – Mar. 2026

- Designed skill-evolution framework for Agents in collaboration with **Adobe Research** using **GRPO** for agentic systems, improving **long-horizon task** solving accuracy on general reasoning benchmarks by 12% (**EMNLP 2026 submission**)
- Researching applicability aware procedural memory for test time inference with **XTrace**, to optimize general reasoning.
- Designing verifiable generative GUI environments to train **Computer Use Agents** with test time control policies to improve agent capability through world model based reasoning.
- Authored a multimodal reasoning benchmark in symbolic music domain with 800+ questions and evaluated multimodal LLMs to identify failure modes in perception based understanding pipelines (**EMNLP 2025**)

Graduate Student Researcher, San Diego Supercomputer Center, CA May. 2025 – Mar. 2026

- Built multi-agent search system over 5M+ datasets by **post training** Qwen3 using DPO for intent identification.
- Developed LLM inference service on **2x A100** gpus using vLLM, DDP and model-distillation cutting p95 latency from 13s → 4s
- Built full-stack app using **MERN** stack to coordinate multi-model LLM workflows, improving retrieval accuracy by 23%

Software Development Engineer, Media.net, India Mar. 2023 – Sep. 2024

- Led an Agile cross-functional team to architect a scalable, real-time bids ranking algorithm using Python, and PostgreSQL, enabling dynamic optimization of advertiser bids, generating over \$2.3M+ in revenue.
- Improved data processing efficiency by building infrastructure in **Go** for data collection, storage, and analysis
- Deployed an anomaly detection system using **PyTorch** to identify bot fraud with 99.3% accuracy

PROJECTS

Visper: Talk to your code – Top 10, Google Gemini Hackathon Oct 2025

- Built real-time voice QA over Github repositories using RAG and Gemini live, assiting in code understanding
- Generated multimodal summaries through Nano Banana and **TTS**, exported guided video walkthroughs

AI Safety Protocol Sep 2025

- Built a code-aware **LLM safety** system for real-time prompt injection, jailbreak, and backdoor attack detection
- Designed multi-agent validation pipelines with real-time telemetry for production AI agents

PUBLICATIONS

A²Mem: Applicability-Aware Procedural Memory for LLM Agents (Under Review) AAAI 2027

Felix Meng, **Yash Vishe**, Hanyu Li, Julian McAuley, Junda Wu

SkillGrid: A Unified Framework for Evaluating Skill Composition in LLM Agents (Under Review) NeurIPS 2026

Gagan Mundada*, Rohan Surana*, **Yash Vishe***, Akshit Agarwal, Xunyi Xiang, Sheldon Yu, Xintong Li, Nikki Lijing Kuang, Jingbo Shang, Julian McAuley, Junda Wu

Skill-R1: Agent Skill Evolution via Reinforcement Learning (Under Review) EMNLP 2026

Yash Vishe, Rohan Surana, Xunyi Jiang, Zihan Huang, Xintong Li, Nikki Lijing Kuang, Tong Yu, Ryan A. Rossi, Jingbo Shang, Julian McAuley, Junda Wu

Evaluating Music Context Preservation: A Framework for Music Editing Systems ISMIR 2026

Yash Vishe*, Eric Xue*, Xunyi Jiang, Zachary Novack, Junda Wu, Julian McAuley, Xin Xu

CSYMR: Benchmarking Compositional Music Reasoning with MIR Tool Integration SIGIR 2026

Boyang Wang, **Yash Vishe**, Xin Xu, Zachary Novack, Julian McAuley, Junda Wu

WildScore: Benchmarking MLLMs in-the-Wild Symbolic Music Reasoning EMNLP 2025

Gagan Mundada*, **Yash Vishe***, Amit Namburi, Xin Xu, Zachary Novack, Julian McAuley, Junda Wu

TECHNICAL SKILLS

Languages: Python, C++, SQL (MySQL, Postgres, SQLite), MongoDB, JavaScript, HTML/CSS

Frameworks: OpenAI GPT, Vertex AI, Pytorch, Langchain, LangGraph, React, Flask, Node.js, Tableau

Developer Tools: Git, GitHub, PySpark, AWS, Docker, Kubernetes, Linux, Google Cloud Platform

Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn